

$$Q(x) = ax^2 \Rightarrow Q''(x) = a$$

DETERMINANT

$$\left[\begin{array}{c|c|c|c} A_1 & A_2 & \dots & A_n \end{array} \right] = A$$

$n \times n$

$$A_i$$

$n \times 1$

$$\det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1 \Rightarrow \det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = -1$$

LPMs

$$|A| = \left| \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix} \right| = a_{11} a_{22} a_{33}$$

$$|A_{12}| = \left| \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} \right| = a_{11} \cdot a_{22}$$

$$|A_1| = |a_{11}| = a_{11}$$

Pf FOR A 2×2 -MAT:

SIGNING QF

$$\text{LET } A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

$$\Rightarrow Q(x) = ax_1^2 + 2bx_1x_2 + cx_2^2$$

$$\begin{array}{l} \text{ADD/SUB} \\ + \frac{b^2}{a} x_2^2 \end{array}$$

$$= \left(ax_1^2 + 2bx_1x_2 + \frac{b^2}{a} x_2^2 \right) + \left(c - \frac{b^2}{a} \right) x_2^2$$

$$= \underbrace{a \left(x_1 + \frac{b}{a} x_2 \right)^2}_{\text{L}} + \underbrace{\left(c - \frac{b^2}{a} \right)}_{\text{L}} \underbrace{x_2^2}_{\text{L}}$$

THE TERM ARE BOTH = 0
ONLY IF $x_1 = x_2 = 0$

$$\begin{array}{l} \bullet a = LPH_1 > 0 \\ \bullet \left(\frac{ca - b^2}{a} \right) = \frac{LPH_2 > 0}{LPH_1 > 0} \end{array} \Rightarrow \begin{array}{l} Q(x) > 0 \\ x \neq 0 \end{array}$$

• FOR PD NEED BOTH COEF $> 0 \Rightarrow LPH_1 > 0$
• FOR ND " " " " $< 0 \Rightarrow LPH_2 > 0$

$$\Downarrow \\ LPH_1 < 0 \\ LPH_2 > 0$$

$$\Rightarrow \begin{array}{l} Q(x) < 0 \\ x \neq 0 \end{array}$$

TAYLOR RESIDUAL

$$\frac{R(h|a)}{h} = \frac{f(a+h) - (f(a) + f'(a)h)}{h}$$

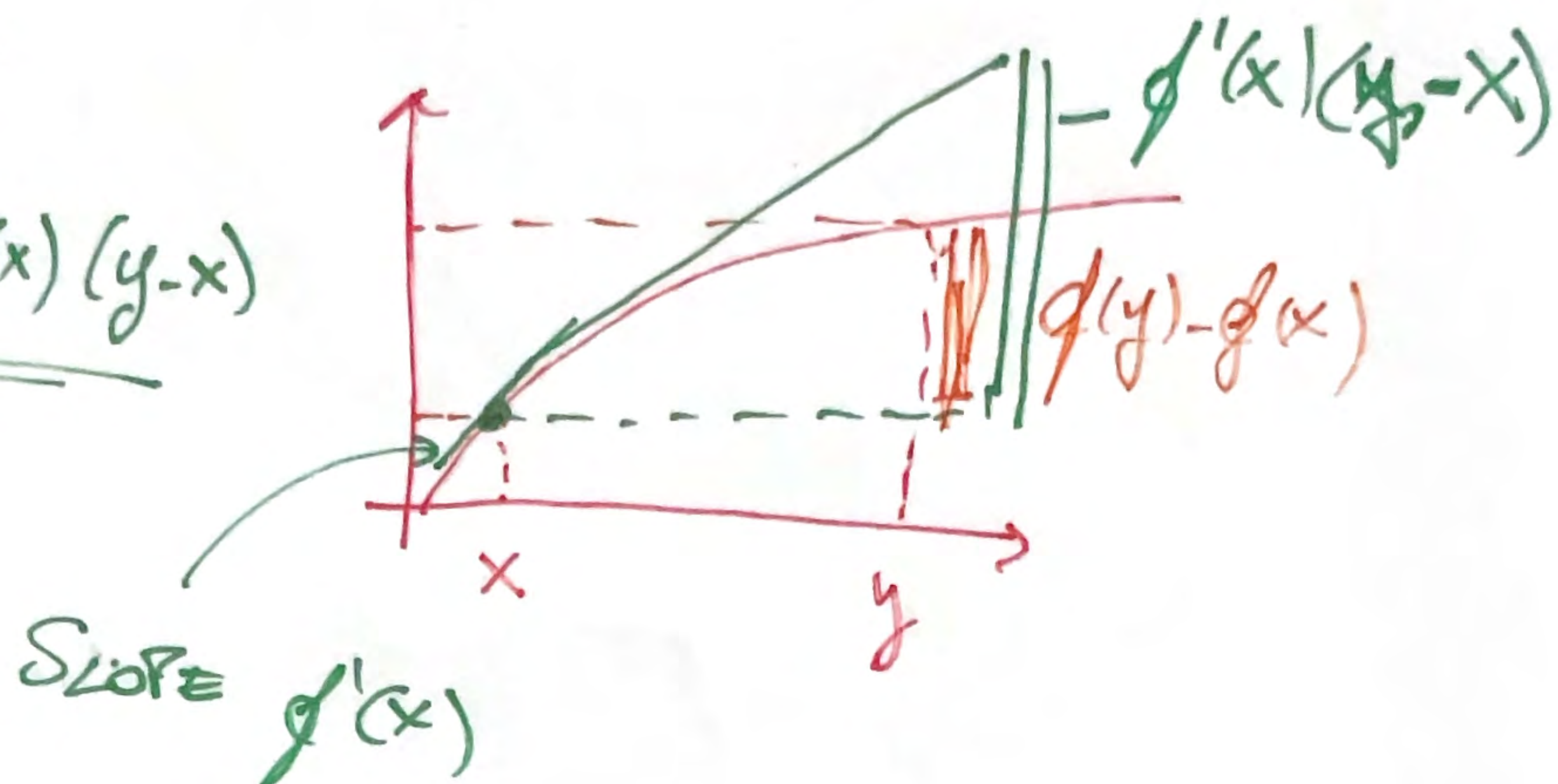
$$= \frac{f(a+h) - f(a)}{h} - f'(a)$$

$h \rightarrow 0$ ↓

$$f'(a) - f'(a) = 0$$

WANT: CONC

$$\underline{\underline{\phi(y) - \phi(x) \leq \phi'(x)(y-x)}}$$



PF 1-DIM: $\Rightarrow$

BY CONC:

$$\phi(ty + (1-t)x) \geq t\phi(y) + (1-t)\phi(x)$$

COLLECT t :

$$\phi(t(y-x) + x) \geq \phi(x) + t(\phi(y) - \phi(x))$$

DIV BY t

& SHIFT

$$\underline{\underline{\phi(x + t(y-x)) - \phi(x) \geq t(\phi(y) - \phi(x))}}$$

MULT & DIV
($y-x$) LHS:

$$\left(\frac{\phi(x + t(y-x)) - \phi(x)}{t(y-x)} \right) (y-x) \geq \quad "$$

$\downarrow t \rightarrow 0$

$$\phi'(x) (y-x) \geq \phi(y) - \phi(x)$$

Pf: $\Leftarrow$ LET $z = ty + (1-t)x$

BY MEAN VALUE DEF:

$$\begin{cases} \phi(y) - \phi(z) \leq \phi'(z)(y-z) \\ \phi(x) - \phi(z) \leq \phi'(z)(x-z) \end{cases} \begin{matrix} t \\ (1-t) \end{matrix}$$

$$t(\phi(y) - \phi(z)) + (1-t)(\phi(x) - \phi(z)) \leq \phi'(z)(ty - tz + (1-t)(x-z))$$

$$\begin{aligned} & \parallel \\ t\phi(y) + (1-t)\phi(x) - \phi(z) & \leq \phi'(z)(\underbrace{ty + (1-t)x}_{=z} - z) = 0 \\ & \quad \quad \quad \underline{\underline{ty + (1-t)x}} \quad \quad \quad \square \end{aligned}$$

RHET CONJ $\Leftrightarrow$

$$f''(x) \leq 0 \quad \forall x$$

PF $\Rightarrow$

BY CONJ

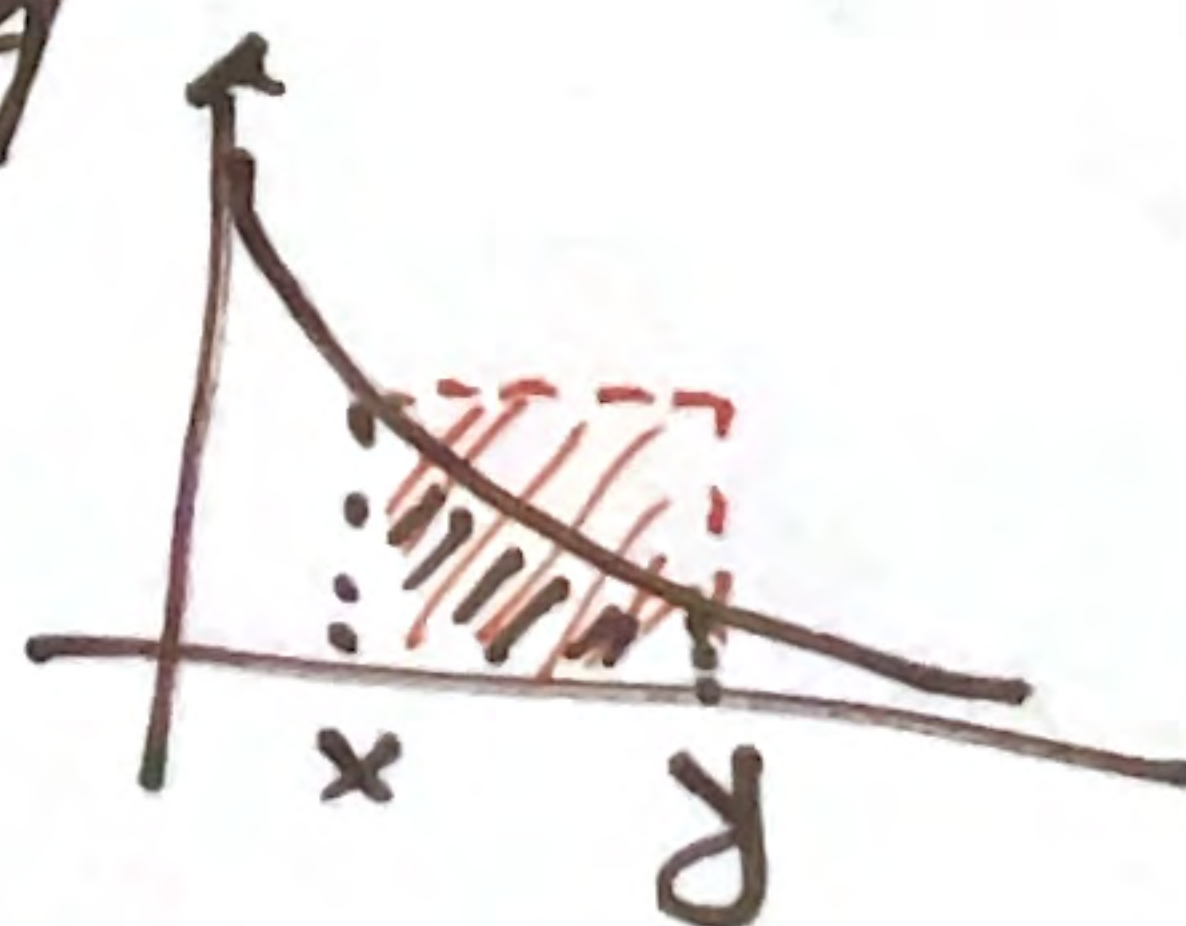
$$f(y) - f(x) \leq f'(x)(y-x)$$
$$f(x) - f(y) \leq f'(y)(x-y) \quad (+)$$

$$0 \leq (f'(x) - f'(y))(y-x)$$

$$\boxed{\begin{array}{l} \text{DIV} \\ -(y-x)^2 \leq 0 \end{array}}$$

$$0 \geq \frac{f'(y) - f'(x)}{y-x}$$

$$\xrightarrow{y \rightarrow x} f''(x) \quad \square$$



PF $\Leftarrow$:

$$f''(x) \leq 0$$

$$\text{BY FTC: } f(y) - f(x) = \int_x^y f'(t) dt$$

$$\leq \int_x^y f'(x) dt = f'(x)(y-x) \quad \square$$

REMARK: ϕ CONC $\Rightarrow$ Q-CONJUNCT

PF : TAKE $y, x \in C_a^+$

$$\Rightarrow \text{BY conc } \phi(ty + (1-t)x) \geq t \underbrace{\phi(y)}_a + (1-t) \underbrace{\phi(x)}_a \geq a$$

$$\Rightarrow ty + (1-t)x \in C_q^+ \quad \square$$

x_1, x_2 APPLY LOG $\Rightarrow \log x_1 + \log x_2$
 x_1, x_2 APPLY EXP $\Leftarrow \log x_1 + \log x_2$
 $\Downarrow$ Q-conv $\Rightarrow$ Q-conv

REMARK: IF f IS Q-conv & $g: \mathbb{R} \rightarrow \mathbb{R}$ $g' > 0$
 $\Rightarrow h(x) = g(f(x))$ IS Q-conv

Pf: IF $f(x) \geq k \iff h(x) \geq g(k)$
, CONVEX

$$\Rightarrow C_x^+(g) = C_{g(x)}^+(h)$$

$$\{x \in U \mid f(x) \geq \kappa\} \quad \{x \in U \mid h(x) \geq g(\kappa)\}$$

REMARK $Q\text{-conc} \iff f(tx + (1-t)y) \geq \min\{f(x), f(y)\}$

PF $\Rightarrow$ TAKE $x, y \in U$ LET $K = \min\{f(x), f(y)\}$

$\Rightarrow x, y \in C_K^+$ BY $Q\text{-CONC}$

$\Rightarrow tx + (1-t)y \in C_K^+$
 $\uparrow$
CONVEX

$\Rightarrow f(tx + (1-t)y) \geq K \quad \square$

PF $\Leftarrow$: TAKE $x, y \in C_K^+$

$\Rightarrow f(tx + (1-t)y) \geq \min\{f(x), f(y)\} \geq K$

$\Rightarrow tx + (1-t)y \in C_K^+ \Rightarrow C_K^+ \text{ IS CONVEX} \quad \square$

REMARK IF $D\phi(x^*)=0$ & $D^2\phi(x^*)$ IS ND

$\Rightarrow x^*$ IS $\leq$ MAX.

PF: TAYLOR:

$$\phi(x^*+h) = \phi(x^*) + \cancel{D\phi(x^*)h} + \frac{h^T D^2\phi(x^*) h}{2}$$

$$+ R_2(h|x^*)$$

$\xrightarrow{\text{SMALL } \|h\|^2}$

$\Rightarrow$ IF h IS SMALL

$$\phi(x^*+h) \leq \phi(x^*) \quad \square$$

IF ϕ IS 1-DIFF & CONCAVE

$$\Rightarrow \phi(y) - \phi(x^*) \leq \underbrace{D\phi(x^*)(y-x^*)}_{=0} \leq 0$$

INTERIOR

① $\Rightarrow \phi(y) - \phi(x^*) \leq 0 \quad \forall y \in \square$

② $\Rightarrow \phi(y) - \phi(x^*) \leq 0 \quad \forall y \in \square$

BOUNDARY

TANGENCY

$$\begin{cases} h(x_1, x_2) = 0 \\ \phi(x_1, x_2) = k \end{cases} \Rightarrow \frac{dx_2}{dx_1} \text{ EQUAL SLOPE}$$

TOT DIFF

$$dx_1 \frac{\partial h(x_1, x_2)}{\partial x_1} + dx_2 \frac{\partial h(x_1, x_2)}{\partial x_2} = 0$$

DIFFERENTIAL AR NUMBERS DIVIDE BY dx_1

$$\Rightarrow \frac{dx_2}{dx_1} \frac{\partial h(x)}{\partial x_2} = - \frac{\partial h(x)}{\partial x_1}$$

$$\Rightarrow \frac{dx_2}{dx_1} = - \frac{\partial h(x)/\partial x_1}{\partial h(x)/\partial x_2}$$

$$\Rightarrow \frac{dx_2}{dx_1} = - \frac{\partial \phi(x)/\partial x_1}{\partial \phi(x)/\partial x_2}$$

$$\mathcal{L}(x|\lambda) = \phi(x) - \lambda h(x)$$

TANGENCY $\Rightarrow$

$$- \frac{\partial h/\partial x_1}{\partial h/\partial x_2} = - \frac{\partial \phi/\partial x_1}{\partial \phi/\partial x_2}$$

$$\left[\begin{aligned} \frac{\partial \phi(x)}{\partial x_1} &= \lambda \frac{\partial h(x)}{\partial x_1} \\ \frac{\partial \phi(x)}{\partial x_2} &= \lambda \frac{\partial h(x)}{\partial x_2} \end{aligned} \right]$$

$$\text{IF } h(x) : \mathbb{R}^n \rightarrow \mathbb{R}$$

$$Dh(x) \neq (0, \dots, 0) \\ 1 \times n$$

$$\text{IF } h : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$Dh(x) \text{ IS FULL RANK} \\ m \times n$$

$$\begin{bmatrix} \partial h_1 / \partial x_1, \dots, \partial h_1 / \partial x_n \\ \partial h_m / \partial x_1, \dots, \partial h_m / \partial x_n \end{bmatrix}$$

$$\|h(x)\|^2 = \underbrace{h(x)^T}_{1 \times m} \underbrace{h(x)}_{m \times 1}$$

$$D\|h(x)\|^2 = \underbrace{2h(x)^T}_{1 \times m} \underbrace{Dh(x)}_{m \times n}$$

$$\lambda(x) = f(x) - \mu g(x)$$

VIOLATE CONSTR $g(x) \not\geq 0$

$$\Rightarrow -\mu g(x) \leq 0 \Rightarrow -\mu \leq 0$$

$$\Rightarrow \mu \geq 0$$

CASE 1

SLACK CONSTRAINT

$$\Rightarrow g(x^*) < 0 \Rightarrow \mu^* = 0$$

$$g(x^*) = 0 \Rightarrow \mu^* \geq 0$$

$$\Rightarrow \begin{cases} g(x^*) \mu^* = 0 \\ \mu^* \geq 0 \\ g(x^*) \leq 0 \end{cases} \quad \text{COMPLEMENT SLACKNESS}$$

$$\text{FOC} \quad Df(x^*) = \mu^* Dg(x^*)$$

FOC

$$\lambda(x|\mu) = 2\sqrt{x} - x - \mu(x-a)$$

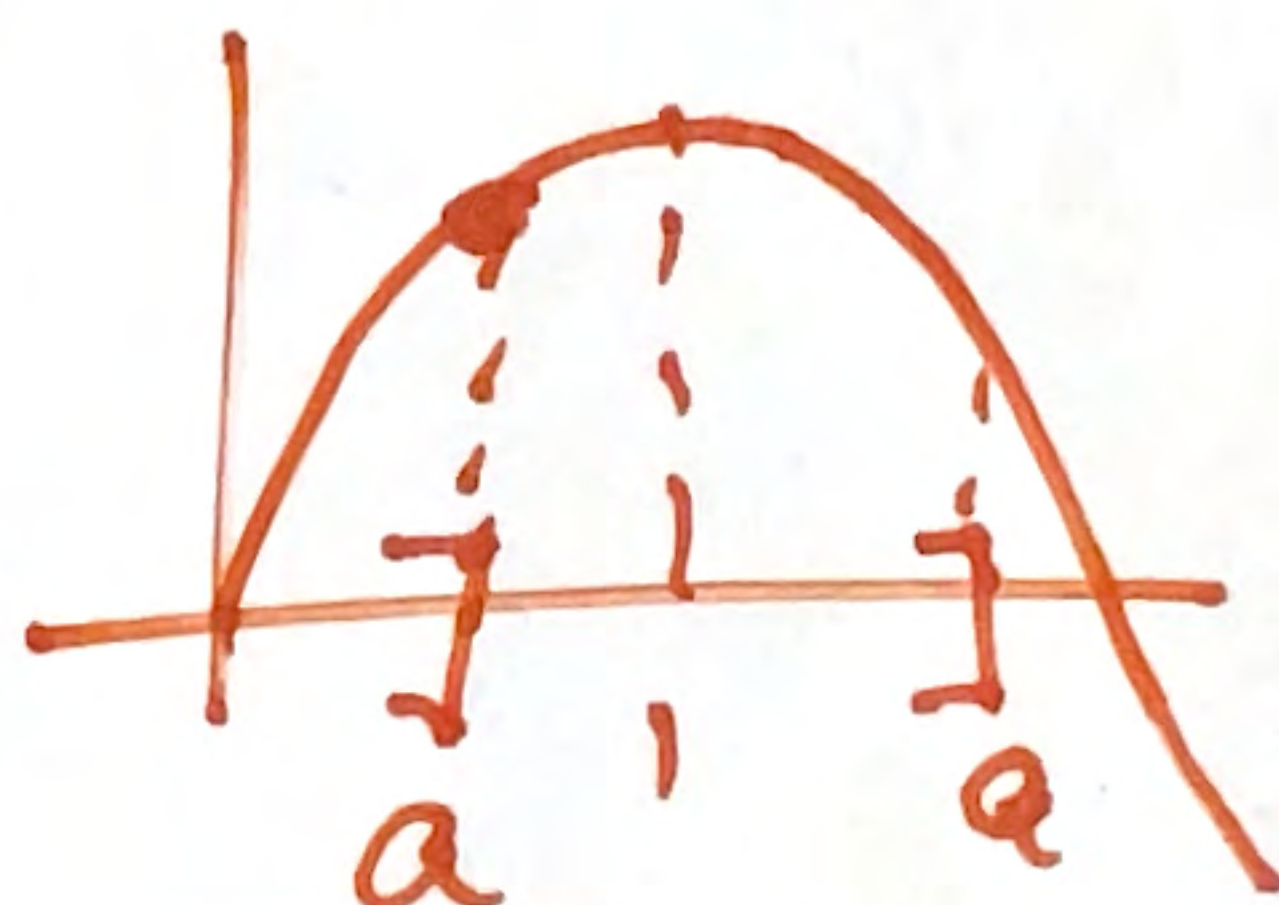
$\Rightarrow$

$$\left\{ \begin{array}{l} \frac{1}{\sqrt{x^*}} - 1 - \mu^* = 0 \checkmark \end{array} \right.$$

$$\mu^*(x^* - a) = 0 \checkmark$$

$$\mu^* \geq 0 \checkmark$$

$$x^* \leq a$$



2 CASES

$$\textcircled{1} \mu^* = 0 \Rightarrow x^* = 1 \leq a$$

$$\textcircled{2} \mu^* > 0 \Rightarrow x^* = a$$

$$\text{FOC} \Rightarrow \mu^* = \frac{1}{\sqrt{a}} - 1 > 0$$

$$\Rightarrow \frac{1}{\sqrt{a}} \geq 1 \Rightarrow a \leq 1$$

SUPPOSE $\mu^* \geq 0$, $g(x^*) \leq 0$, $\mu^* g(x^*) = 0$

& x^* MAXES $\mathcal{L}(x|\mu^*)$

$\Rightarrow x^*$ SOLUTION...

PF: $\underset{\uparrow}{\mathcal{J}(x^*)} = \underset{\uparrow}{\mathcal{J}(x^*)} - \mu^* g(x^*) = \mathcal{L}(x^*|\mu^*)$

BY CS...

$\underset{\uparrow}{\geq} \mathcal{L}(x|\mu^*) = \mathcal{J}(x) - \mu^* g(x)$
AS x^* MAXES $\mathcal{L}$

$\underset{\uparrow}{\geq} \mathcal{J}(x)$ $\square$

IF x IS S.T. $g(x) \leq 0$

$\Rightarrow -\mu^* g(x) \geq 0$

IF $\mathcal{L}$ IS CONV IN $x \Rightarrow$

$D_x \mathcal{L}(x^*|\mu^*) = 0$ IS SUFF
FOR OPT...

$D_x \mathcal{J}(x) = \mu^* D_x g(x^*)$ BY UNCONST RES

$\mathcal{L}$ IS CONV IF $\mathcal{J}$ IS CONV & g_i IS CONV $\forall i$

PF: ENVELOPE THM

$$D_x = \frac{\partial}{\partial x}$$

$$D_b = \frac{\partial}{\partial b}$$

$$\frac{dv(b)}{db} = \frac{d \phi(x(b)|b)}{db}$$

$\begin{matrix} & \textcircled{2} & \textcircled{1} \\ & \swarrow & \swarrow \\ & \phi(x(b)|b) & \end{matrix}$

$$= D_b \phi(x(b)|b) + D_x \phi(x(b)|b) D_b x(b) =$$

$\begin{matrix} 1 \times 1 & & 1 \times n & n \times 1 \\ & \text{BY FOC} & & \end{matrix}$

$$= \text{"} + \underbrace{\mu(b) D_x \phi(x(b)|b)}_{\text{L}^*} D_b x(b)$$

$$= D_b \phi(x(b)|b) - \mu(b) D_b \phi(x(b)|b)$$

$$= D_b \mathcal{L}(x(b), \mu(b)|b) \quad \square$$

RESTRICT TO $g(x(b)|b) = 0$ BY CS & $\mu(b) \rightarrow 0$

TOT DIFF WRT b

$\begin{matrix} & \textcircled{2} & \textcircled{1} \\ & \swarrow & \swarrow \\ & g(x(b)|b) & \end{matrix}$

L^*

$$\Rightarrow D_b g(x(b)|b) + D_x g(x(b)|b) D_b x(b) = 0$$

IFT: APPLICATION

$$F(x(b) | b) = 0$$

TOT DIFF WRT b_i (1)

b CLOSE TO $\bar{b}$

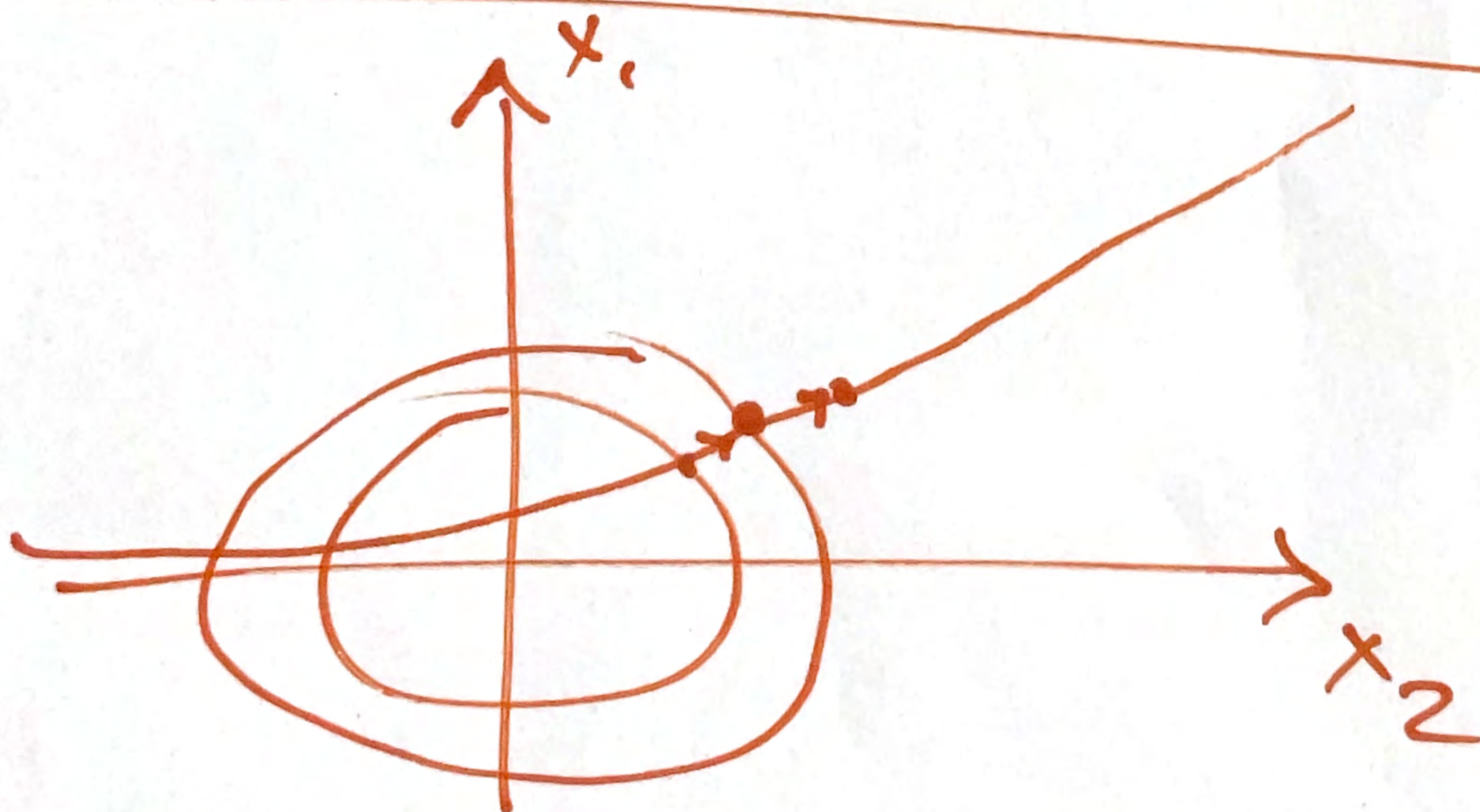
WANT

$$D_{b_i} F(x(b) | b) + D_x F(x(b) | b) D_{b_i} x(b) = 0$$

$N \times 1$ $N \times N$ $N \times 1$

$\Rightarrow$
AS DET $\neq 0$

$$D_{b_i} x(b) = -[D_x F(x(b) | b)]^{-1} D_{b_i} F(x(b) | b)$$



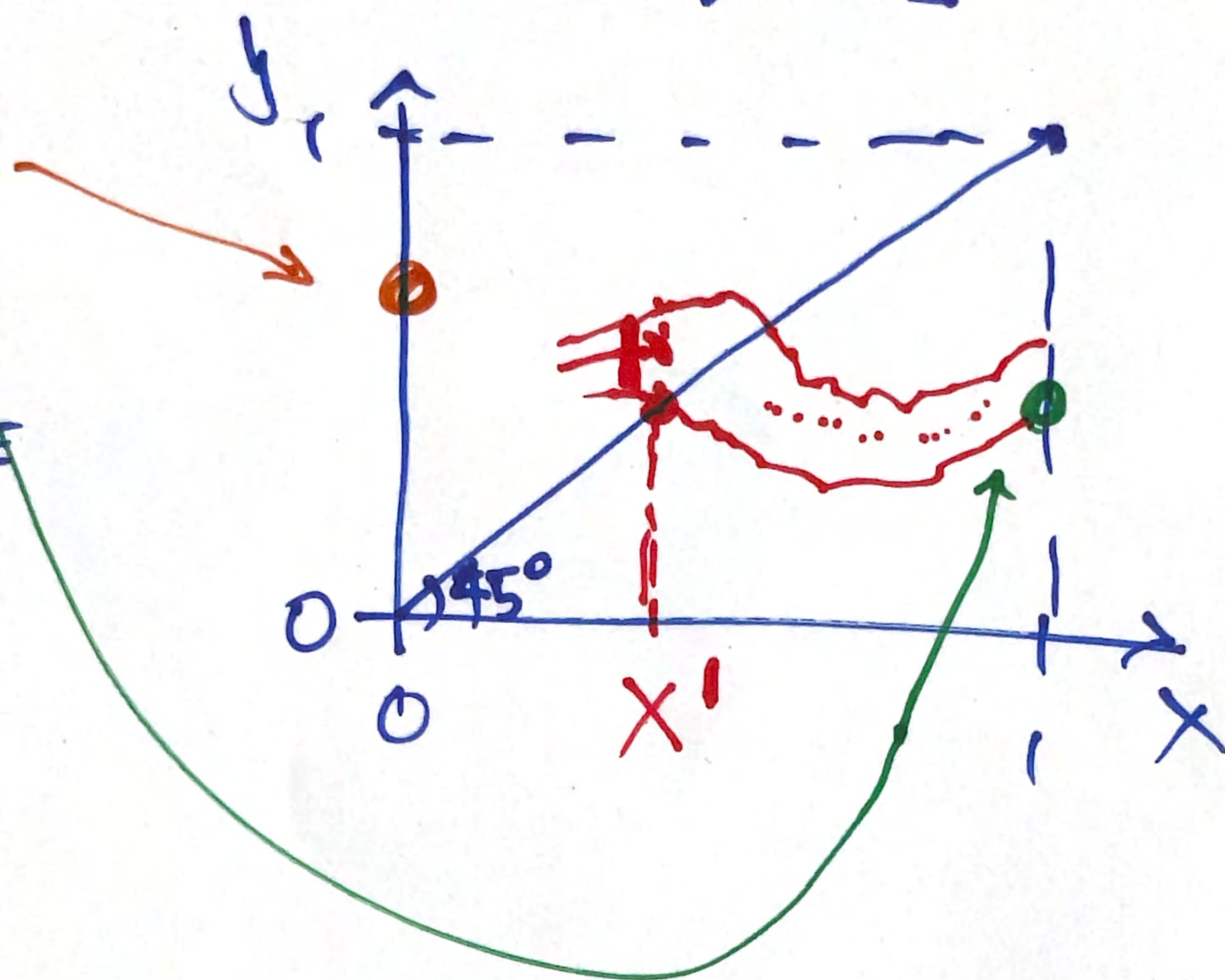
$$F(x) = 0 \iff x = \underbrace{x + F(x)}_{H(x)}$$

SUPPOSE $h: [0,1] \Rightarrow [0,1]$

$0 \notin h(0)$

$1 \notin h(1)$

OTHERWISE
I'M DONE



SMALLEST x ST $\exists y \leq x$

~~$h(x) \geq y$~~

$\Rightarrow x < x' \Rightarrow y \in h(x) \Rightarrow y > x$

$\Rightarrow$ BY UHC $\exists y' \in h(x')$ ST $y' \geq x'$

BUT $\exists y'' \in h(x')$ ST $y'' \leq x'$

BUT BY CONVEX $\forall t y' + (1-t)y'' \in h(x')$
" " "
 x' FOR SOME t